Foundations of Mathematics

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Motivation: there are lots of proof assistants which allow to prove mathematical theorems, but their use for the purposes of an average mathematician is deeply disappointing due to various reasons.

I am creating a proof assistant and this document is a very early prologue to what the computer-generated article will look like.

The other point of creating such document is to provide programmed classical logical foundations for developing a set-theoretic proofs, and vice-versa to provide a set-theoretic foundations to prove theorems about logic.

0.1 Goals

Despite theory being presented is sometimes unusually presented, my intention was to keep it as standard as possible. Any "extravagant" part has reason behind it: for example, it is absurd to speak about truthness of the statements $f(x) = 5$, when $x \notin \mathrm{dom}(f)$. We may want to say that it is false, but then it creates the assymetry: $f(x) \neq 5$ means that some of two cases is true: 1) $x \notin \mathrm{dom}(f)$ or 2) $((x \notin \mathrm{dom}(f))$ and f(x) is actually equal to something else). But when we have 3-valued logic, then we get rid of such compromises. In it $(f(x) \neq 5)$ means that both sides are defined and they are not equal.

- The aim of this document is to provide standard classical mathematical foundations as well as some amount of perfect proofs.
- The proof assistant based on this very classical theory is being implemented. `https://github.com/georgydunaev/set_theory_prover`
- After completion of development of the program, not only tactics, but also machine learning algorithms may be applied in a search of the proofs.